

EXPRIMENT-5

5. Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p , substitute the ciphertext letter C : $C = E([a, b], p) = (ap + b) \bmod 26$. A basic requirement of any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k, p) \neq E(k, q)$. Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a . For example, for $a = 2$ and $b = 3$, then $E([a, b], 0) = E([a, b], 13) = 3$.

- a. Are there any limitations on the value of b ?
- b. Determine which values of a are not allowed.

PROGRAM:

```
1
2 import math
3
4 plaintext = input("Enter the plaintext: ").upper()
5 a = int(input("Enter the value of a: "))
6 b = int(input("Enter the value of b: "))
7
8 if math.gcd(a, 26) != 1:
9     print("Invalid value of a")
10 else:
11     ciphertext = ""
12
13     for ch in plaintext:
14         if ch.isalpha():
15             p = ord(ch) - ord('A')
16             c = (a * p + b) % 26
17             ciphertext += chr(c + ord('A'))
18         else:
19             ciphertext += ch
20
21     print("Ciphertext:", ciphertext)
22
```

OUTPUT:

```
Enter the plaintext: hello
Enter the value of a: 5
Enter the value of b: 3
Ciphertext: MXGGV
```